

CCAT & CCEE Deployment Manual

Prerequisites:

Ensure that you have two latest laptops. One of these laptops will act as **PRIMARY** server, the other one will be used as **SECONDARY** server.

Custom OS Installation and Configuration

Custom OS is based on Centos 9 Stream. Link to download Custom OS iso image will be shared separately via Email/WhatsApp.

STEP 1:

Download the Custom OS iso image and make it as **BOOTABLE PENDRIVE** using **rufus** software. Then install the Custom OS on both servers.

STEP 2:

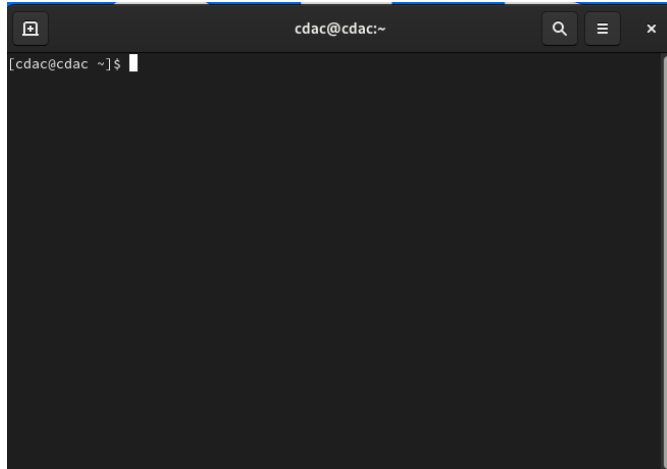
After OS installation, Login as “cdac” user (default password “cdac123”).



STEP 3:

Now kindly set the hostname by following command:

Open **terminal**:



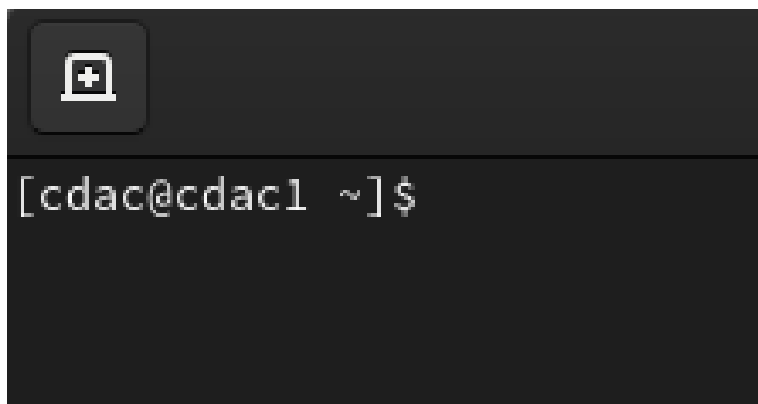
\$ sudo hostnamectl set-hostname cdac1 (on PRIMARY server)

Close the terminal.

\$ sudo hostnamectl set-hostname cdac2 (on SECONDARY server)

Close the terminal.

Now, whenever you open a new terminal, you will see the effect, new hostname will be appeared on terminal screen.



STEP 4:

Please ensure that you set **STATIC IP** address on both servers.

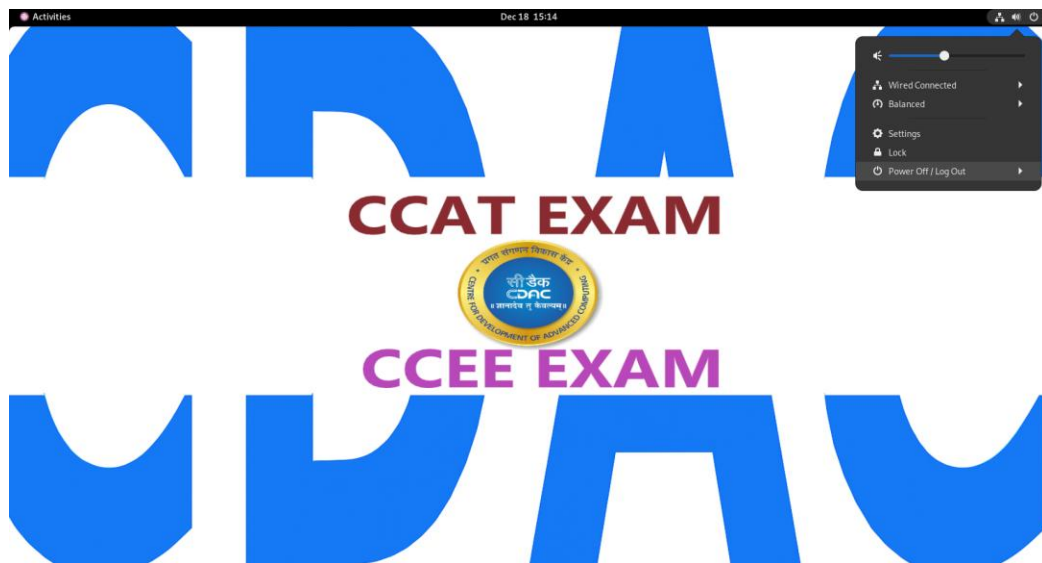
Static IP for PRIMARY:

IP: 10.0.0.1, NetMask: 255.255.0.0 and Gateway 0.0.0.0.

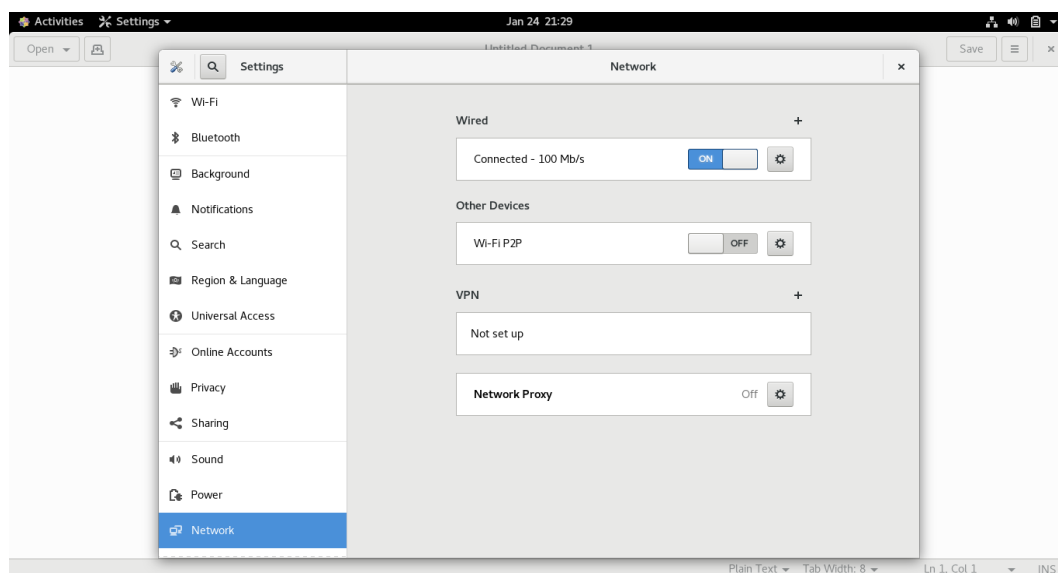
Static IP for SECONDARY:

IP: 10.0.0.2, NetMask: 255.255.0.0 and Gateway 0.0.0.0.

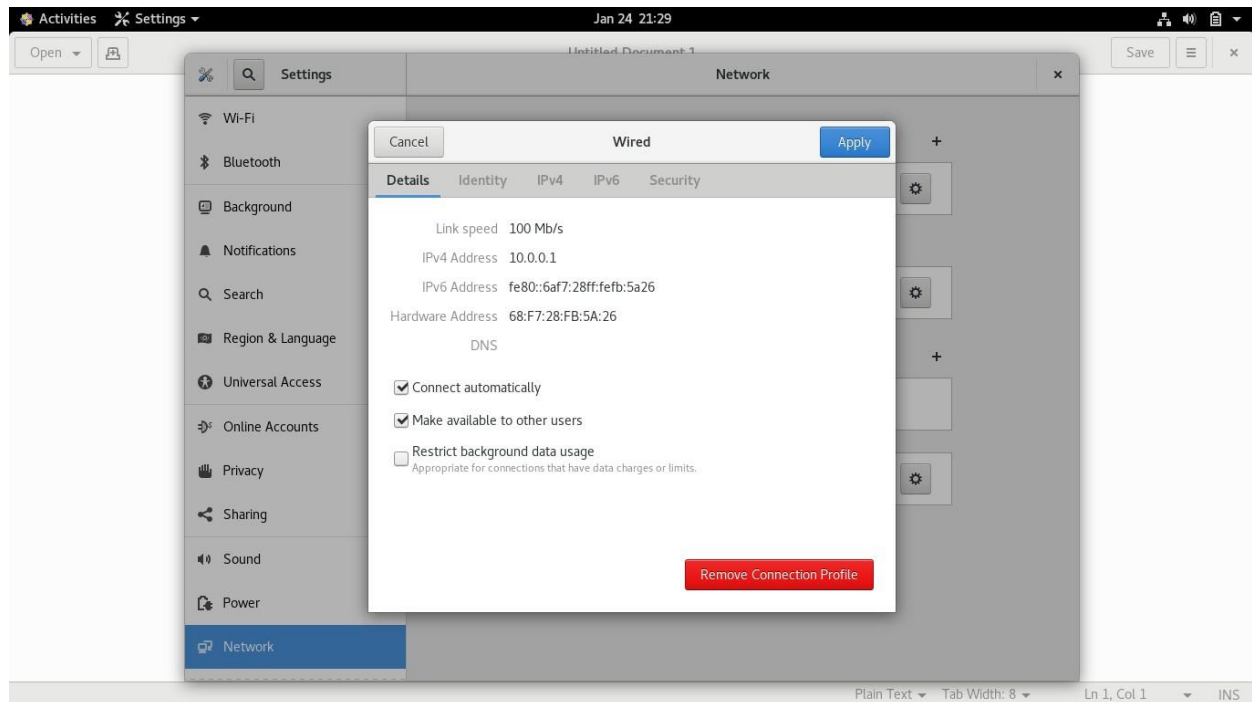
Click on network icon provided on the upper right corner of screen, as given in the picture below:



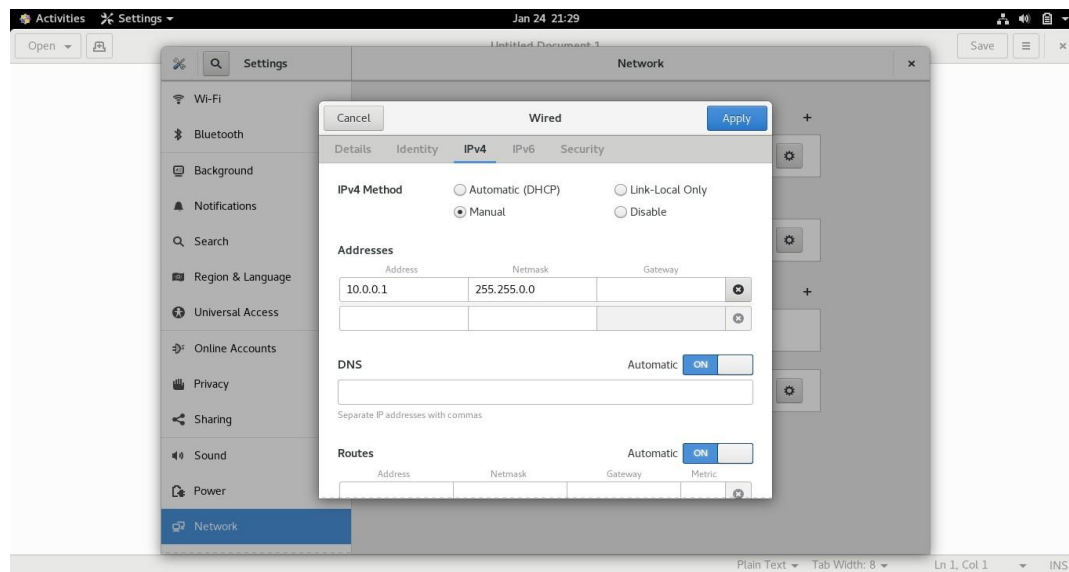
After clicking on network icon, click on setting:



In the above given figure select the option 'Network' and then click on the setting icon of the 'Wired' option. Clicking the same will display the following window:



In the above given figure check the option 'Connect Automatically' then select IPV4 setting tab and select the method as manual and enter the IP Details as shown in the below figure:



STEP 5:

Ensure that both servers and candidate PCs are connected in the same network. Ensure that these two servers can ping each other.

The following command can be used to ping systems on the Terminal:
ping <IP Address>

\$ ping 10.0.0.2 (on PRIMARY server)

\$ ping 10.0.0.1 (on SECONDARY server)

NOTE:

STEP 1 to 5 needs to be performed on both the primary and secondary servers.

First Time Server Deployment Process

PLEASE ENSURE THAT YOU HAVE DONE ALL THE STEPS MENTIONED IN PREREQUISITES SECTION ABOVE.

STEP 1:

Copy all following files in /home/cdac/Downloads folder of both servers.

1. EXAM-DeploymentPackage.tar.gz
2. untar.sh
3. Cleanup.sh
4. source_exam_setup.sh
5. BackupEXAM.sh

STEP 2:

Ensure that both the servers are connected with each other before running any script.

Set the time on primary and secondary servers using the following command

Ex. `$ sudo date -s "dd Mon YYYY HH:MM:SS +5:30"`

i.e.: `$ sudo date -s "11 JAN 2026 09:00:00 +5:30"` (PRIMARY & SECONDARY)

Ensure that the two laptops should be in sync with respect to time with each other.

STEP 3:

Run the following commands on both PRIMARY as well as SECONDARY servers:

`$ cd ~/Downloads` (PRIMARY & SECONDARY)

`$ chmod 777 *` (PRIMARY & SECONDARY)

STEP 4:

After running the above commands, run the following script only on Primary Server and please ensure that all the terminal are closed on the secondary server:

`$ source source_exam_setup.sh` (PRIMARY Server Only)

Running the above command will ask for the IP and hostname of the Primary and Secondary server. Enter the details as prompted carefully.

After running above script, deployment will be done on both the servers.

CI Activities

All CI activities should be performed on secondary server only (recommended).

Open the Firefox browser and check the Chief Invigilator (CI) login:

CI Login Link:

<https://<PRIMARY-IP>:8443/onlineexam/login>

i.e. **<https://10.0.0.1:8443/onlineexam/login>**

Default CI User Credentials: Login ID - **csuser**, Password - **Test@123**

Student Login Link:

<https://<PRIMARY-IP>:8443/onlineexam/exam/login>

i.e. **<https://10.0.0.1:8443/onlineexam/exam/login>**

BACKUP Process

Take backup after each slot and in the end of examination. Take backup of **RESPONSE SHEETS** generated through the exam software also.

Backup can be taken by executing the following commands on the terminal:

```
$ cd ~/Downloads (PRIMARY & SECONDARY)  
$ ./BackupEXAM.sh (PRIMARY & SECONDARY)
```

Backup folder will be created in **EXAM-Backup folder at /home/cdac**. It will also contain center code in the name. Backup folders will be required after the examination.

NOTE:

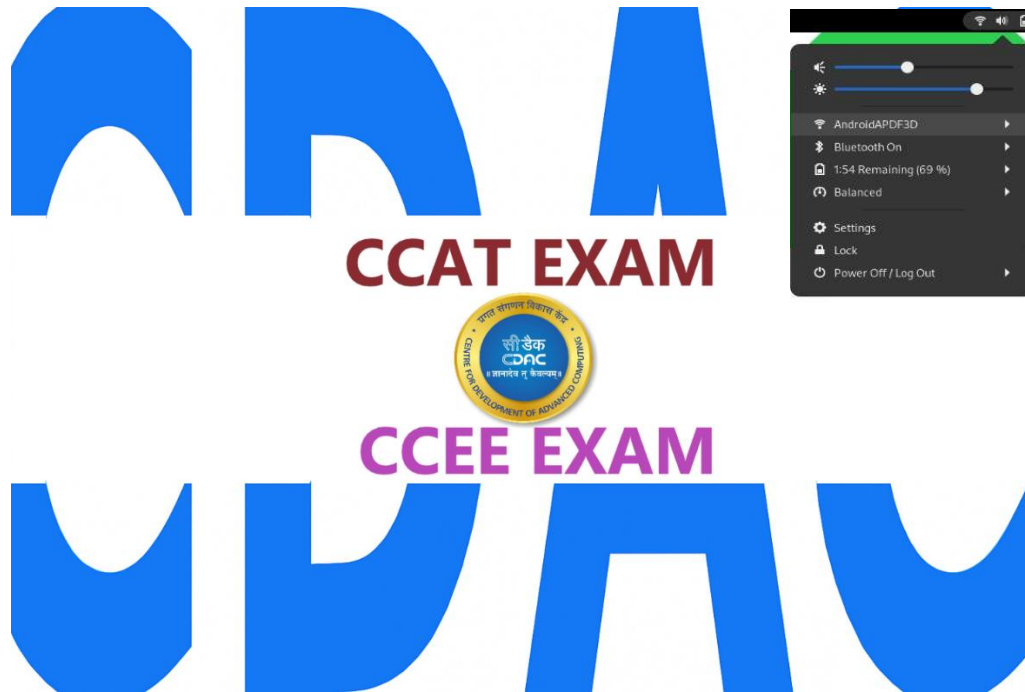
On the day of actual examination, please don't delete these folders and take the backup of these folders after every slots.

It is to be ensured that backup folder is created under location /home/cdac/EXAM-Backup after performing the backup activity and before starting the next slot.

Allow Remote Access

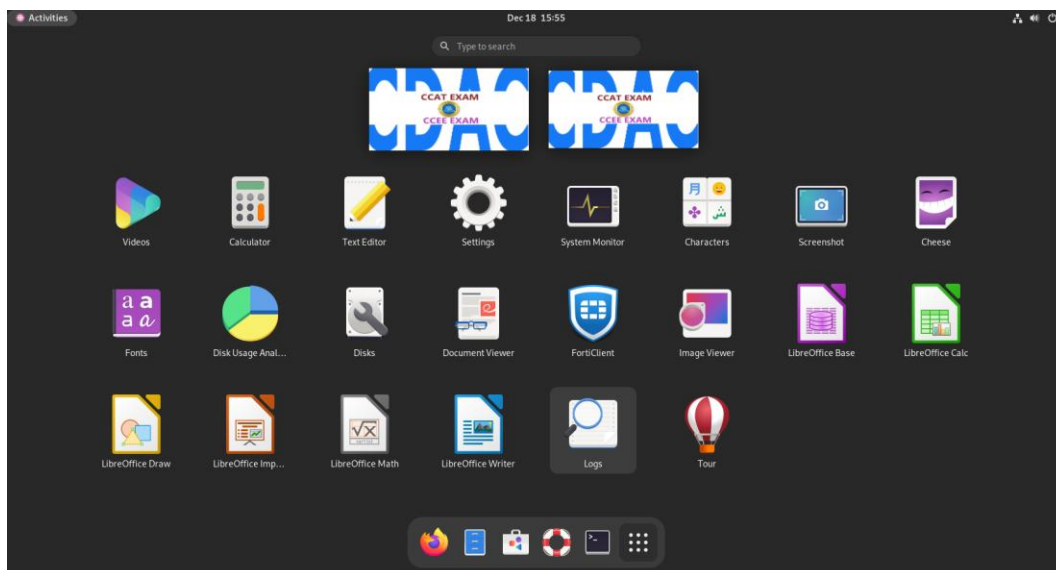
STEP 1:

Connect the server machine to the internet through **Wi-Fi**.



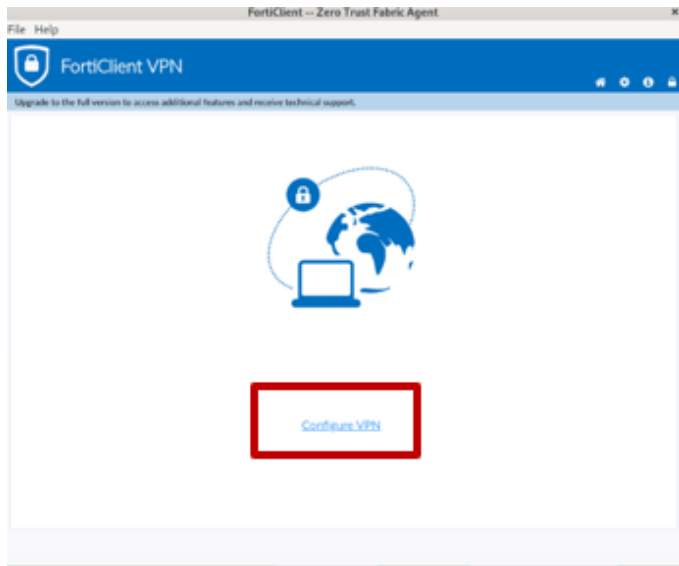
STEP 2:

Go to the activities at the upper left corner and open **FortiClient** application.



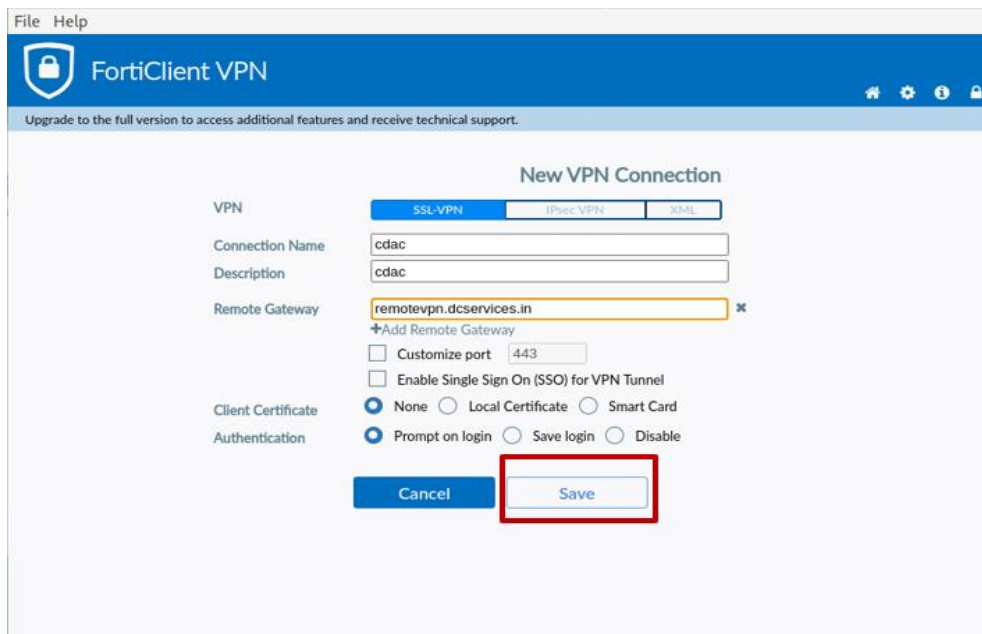
STEP 3:

For initial setup, Click on **Configure VPN**.



STEP 4:

After clicking on the **configure VPN**, given below screen will be displayed.
In the given screen, select **SSL-VPN** and enter the following -
Connection Name: **cdac**
Description: **cdac**
Remote Gateway: **remotevpn.dcservices.in**
and click on '**save**' button to save all the settings.

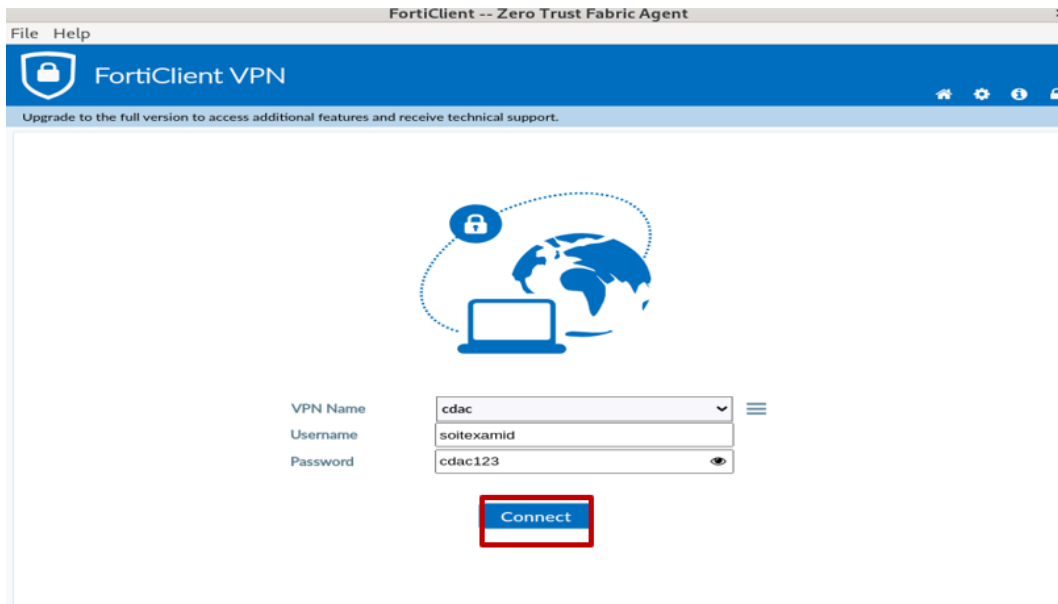


STEP 5:

Now in the **cdac VPN name** tab, enter the following -

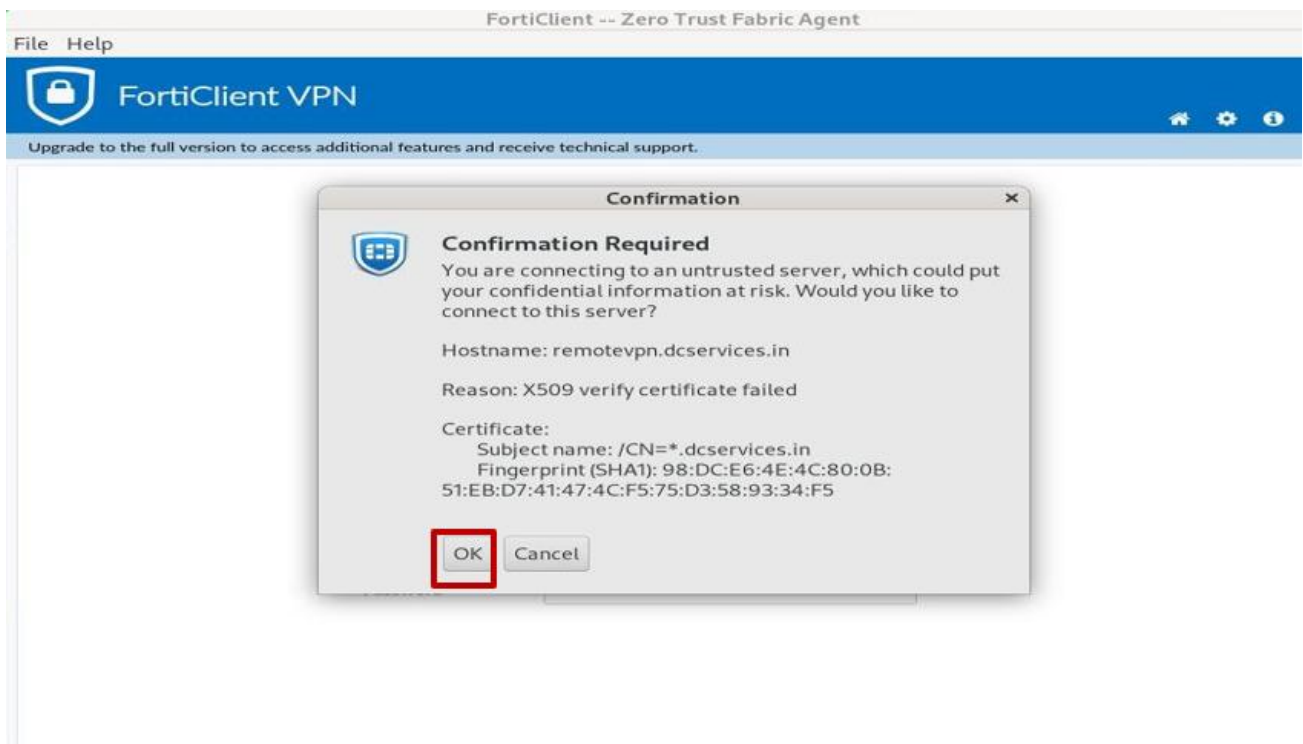
Username: **soitexamid**

Password: **cdac123**



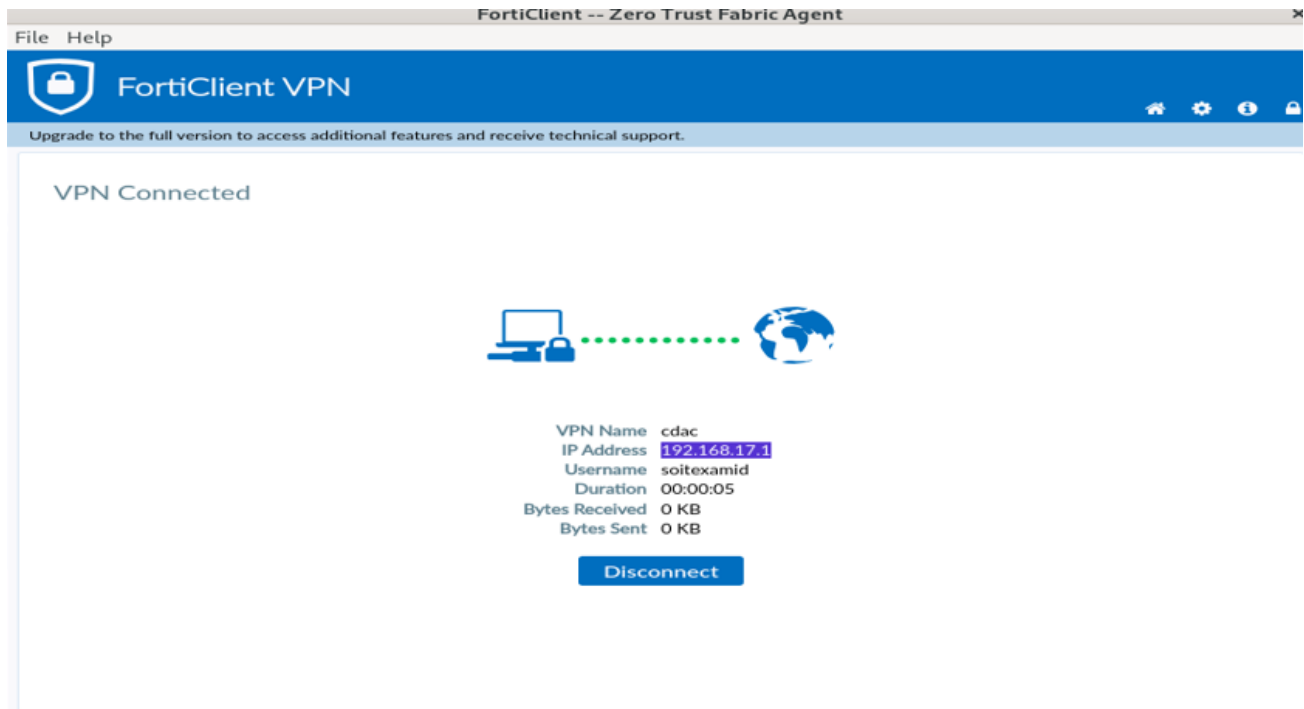
STEP 6:

After entering the login credentials, click on 'Connect', then a confirmation popup window will appear, then click on 'OK'.



STEP 7:

Once the VPN connected, an IP Address will be generated for your VPN connection. CI needs to communicate this IP to helpdesk in order to provide the remote access to the CDAC NOIDA helpdesk team members.



STEP 8:

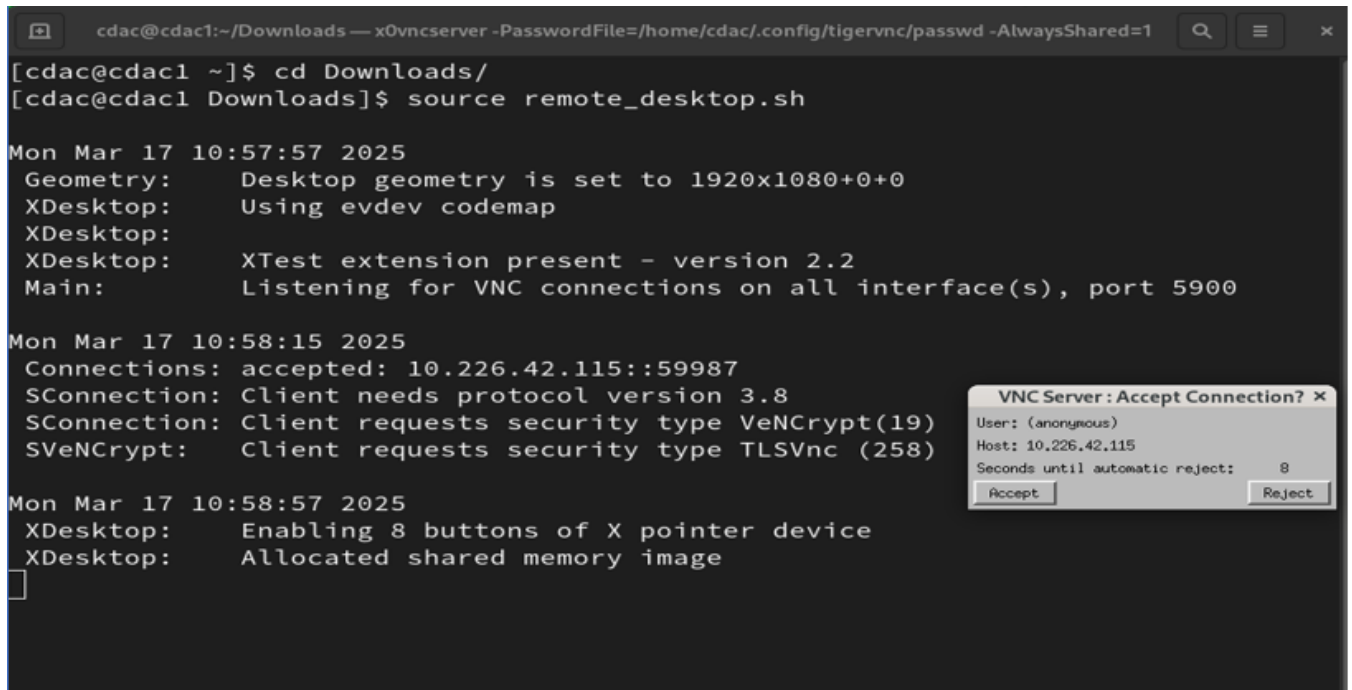
Before providing the IP to the helpdesk, open the terminal on server and move to the download directory and run the command `source remote_desktop.sh`.

```
cdac@cdac1:~/Downloads — x0vncserver -PasswordFile=/home/cdac/.config/tigervnc/passwd -AlwaysShared=1
cdac@cdac1 ~]$ cd Downloads/
cdac@cdac1 Downloads]$ source remote_desktop.sh

Mon Mar 17 10:57:57 2025
Geometry:   Desktop geometry is set to 1920x1080+0+0
XDesktop:   Using evdev codemap
XDesktop:
XDesktop:   XTest extension present - version 2.2
Main:       Listening for VNC connections on all interface(s), port 5900
```

STEP 9:

Once the helpdesk member will try to take the remote access Ci has to accept the connection.



The screenshot shows a terminal window titled "cdac@cdac1:~/Downloads — xOvncserver -PasswordFile=/home/cdac/.config/tigervnc/passwd -AlwaysShared=1". The terminal output shows the user navigating to the Downloads directory and sourcing the remote_desktop.sh script. The script sets desktop geometry to 1920x1080, uses evdev codemap, and listens for VNC connections on port 5900. A connection is accepted from 10.226.42.115. The terminal also shows the client requesting protocol version 3.8 and security types VeNCrypt(19) and TLSVnc (258). A dialog box titled "VNC Server : Accept Connection? x" is overlaid on the terminal, showing the user as (anonymous), host as 10.226.42.115, and a countdown of 8 seconds until automatic reject. The dialog has "Accept" and "Reject" buttons.

```
cdac@cdac1:~/Downloads — xOvncserver -PasswordFile=/home/cdac/.config/tigervnc/passwd -AlwaysShared=1
[cdac@cdac1 ~]$ cd Downloads/
[cdac@cdac1 Downloads]$ source remote_desktop.sh

Mon Mar 17 10:57:57 2025
Geometry:      Desktop geometry is set to 1920x1080+0+0
XDesktop:      Using evdev codemap
XDesktop:
XDesktop:      XTest extension present - version 2.2
Main:          Listening for VNC connections on all interface(s), port 5900

Mon Mar 17 10:58:15 2025
Connections:   accepted: 10.226.42.115::59987
SConnection:   Client needs protocol version 3.8
SConnection:   Client requests security type VeNCrypt(19)
SVeNCrypt:     Client requests security type TLSVnc (258)

Mon Mar 17 10:58:57 2025
XDesktop:      Enabling 8 buttons of X pointer device
XDesktop:      Allocated shared memory image

```

NOTE:

In case of facing any issues before/during/after examination, kindly contact **NOIDA HELP DESK** Team immediately.

THANK YOU.